https://doi.org/10.1038/s44298-024-00059-9

Multivalent interactions between fully glycosylated influenza virus hemagglutinins mediated by glycans at distinct N-glycosylation sites

Check for updates

Ruofan Li[1,2,3,7], Jingjing Gao[1,5,7], Lin Wang[1,6], Miao Gui[4] ✉ & Ye Xiang[1,2,3] ✉

The hemagglutinin (HA) glycoprotein of influenza virus binds host cell receptors and mediates viral entry. Here we present cryo-EM structures of fully glycosylated HAs from H5N1 and H5N8 influenza viruses. We find that the H5N1 HA can form filaments that comprise two head-to-head HA trimers. Multivalent interactions between the two HA trimers are mediated by glycans attached to N158. The distal Sia1-Gal2-NAG3 sugar moiety of N158 interacts with the receptor binding site on the opposing HA trimer. Additional interactions are observed between NAG3 and residues K222 and K193. The H5N8 HA lacks the N158 glycosylation site and does not form the filamentous structure. However, the H5N8 HA exhibits an auto-inhibition conformation, where the receptor binding site is occupied by the glycan chain attached to residue N169 from a neighboring protomer. These structures represent native HA-glycan interactions, which may closely mimic the receptor-HA interactions on the cell surface.

Influenza A virus (IAV) belongs to the *Orthomyxoviridae* family and is the major cause of seasonal and pandemic flu. The polymorphic IAV particles have a membrane envelope that encapsulates eight segmented negative RNA genomes[1]. IAV infects mammals as well as birds[2] and replicates mainly in the host intestinal tract or upper respiratory tract[3,4]. Hemagglutinin (HA) and neuraminidase (NA) are the two major surface proteins of IAV. HA is responsible for receptor binding and for mediating membrane fusion during virus entry and is the major target recognized by the immune system[5]. NA is responsible for virion release and plays a role in initiating virus binding and entry as having been indicated by recent studies[6–8]. The molar ratio of HA and NA on the surface of the virions is estimated to be 5:2[9]. Cross-species infection of IAV is considered the major cause of pandemic flu. Surface antigens of IAVs that infect different species are significantly different and are strong stimulators to the immune system of the newly adapted hosts[10].

Binding of HA to the sialic acid (SA) receptors on the host cell surface is the first step of influenza virus entry. The SA receptors are complex glycans that have an SA residue at the distal end and are covalently connected to cell surface proteins or probably lipids. For avian influenza viruses, HA prefers binding the α-2, 3 sialylated glycans in a cone-like topology. Human-adapted influenza viruses bind specifically to the long-branched α-2, 6 sialylated glycans in either the cone-like or the umbrella-like topology[11,12]. Complex structures of HA and the receptor analogous glycan LSTa (α-2, 3 linked lactoseries tetrasaccharide, Sia-α2, 3-Gal-β1, 3-NAG-β1, 3-Gal-β1, 4-Glc) or LSTc (α-2, 6 linked lactoseries tetrasaccharide, Sia-α2, 6-Gal-β1, 4-NAG-β1, 3-Gal-β1, 4-Glc) revealed the detailed interactions between the terminal SA and the HA receptor binding site (RBS)[13]. Host SA receptors are a determinant factor of influenza virus tropism. The α-2, 6 sialylated glycans recognized by human influenza A viruses are rich on the surface of the epithelial cells in the human upper respiratory tract. The α-2, 3 sialylated glycans, which are specifically recognized by the HA of avian IAV, are rich on the surface of epithelial cells in the human lower respiratory tract and are barely accessible by the viruses[14]. The highly pathogenic avian IAV H5N1 has only limited transmission efficiency in human hosts. However, once the mutant HA (N158D/N224K/Q226L/T318I or H110Y/T160A/Q226L/G228S) of H5N1 is adapted to bind the α-2, 6 sialylated glycans, efficient airborne transmission of H5N1 could be enabled among ferrets[15,16].

Binding of the HA molecule to a single sialylated glycan is weak and multivalent interactions could be critical for efficient binding and

[1]Beijing Frontier Research Center for Biological Structure, Center for Infectious Disease Research, School of Basic Medical Sciences, Tsinghua University, 100084 Beijing, China. [2]SXMU-Tsinghua Collaborative Innovation Center for Frontier Medicine, Shanxi Medical University, 030001 Taiyuan, Shanxi Province, China. [3]Tsinghua-Peking Center for Life Sciences, 100084 Beijing, China. [4]Department of Obstetrics and Gynecology, Zhejiang Key Laboratory of Precise Protection and Promotion of Fertility, Sir Run Run Shaw Hospital, Zhejiang University School of Medicine and Liangzhu Laboratory, 310016 Hangzhou, China. [5]Present address: Department of Cell Biology, Harvard Medical School, Boston, MA, 02115, USA. [6]Present address: Guangzhou National Laboratory, 510320 Guangzhou, China. [7]These authors contributed equally: Ruofan Li, Jingjing Gao. ✉e-mail: miaogui@zju.edu.cn; yxiang@mail.tsinghua.edu.cn

subsequent virus entry into host cells[17,18]. The binding mode of HA to the glycan moiety on protein or lipid is not clearly defined. In addition, the exact functions of the glycans attached to HA have not been clearly understood. Here we report the cryo-EM structures of two recombinant HAs with complex glycan moieties. For the HA from H5N1, a large portion of the HAs form filaments through glycan-mediated multivalent interactions, which may represent the multivalent binding mode of the HA to cell surface receptors. We also observed in the HA of H5N8 that the receptor binding pocket is completely blocked by the glycan moiety attached to a neighboring protomer.

Results

Production of the glycosylated H5N1 and H5N8 HAs

Compared to the H3-HA, the H5 subtype of HA (H5-HA) lacks the N165 glycosylation site on the head but contains two additional glycosylation sites at N158 and N169. The N158 glycosylation site is commonly found in H5-HAs of avian influenza viruses that emerged during 2004–2005[19], such as the A/Vietnam/1203/2004 strain, which was first isolated from a human patient in 2004 and belongs to clade 1. However, as H5 avian influenza viruses continued to evolve, the N158 glycosylation site was eventually lost in certain strains. To investigate the impact of changes in glycosylation sites on HA during the virus's evolution, we selected the HAs of two strains for study and comparison: A/Vietnam/1203/2004 (H5N1) strain, which processes the N158 glycosylation site, and A/chicken/Czech_Republic/1566-1/2021 (H5N8), a strain from clade 2.3.4.4b that lacks the N158 glycosylation site.

We expressed the full-length HA of the H5N1 avian influenza virus (A/Vietnam/1203/2004, H5N1) (H5N1-HA) and the ectodomain (residues 1-526) of the avian IAV H5N8-HA (A/chicken/Czech_Republic/1566-1/2021, H5N8) in fusion with a C-terminal strep tag using HEK293F cells. To better mimic the behavior of native HA, stable H5N1-HA trimers were obtained without any trimerization tags, which are frequently used for the HA ectodomain expression[20]. The HAs produced in HEK293F are highly glycosylated, as indicated by the smear bands of the HA0, HA1 and HA2 on the SDS-PAGE gels (Supplementary Fig. 1a). Most of the precursor HA0 molecules were cleaved into HA1 and HA2 (Supplementary Fig. 1a). Size exclusion chromatography (SEC) analysis of the H5N1-HA showed two major peaks. The earlier elution peak at 12.56 ml from the Superose-6 column has an apparent molecular weight of ~400 kDa, which corresponds to a HA pentamer or hexamer (Supplementary Fig. 1a). The later elution peak at 14.25 ml has an apparent molecular weight of ~200 kDa, which corresponds to a HA trimer (Supplementary Fig. 1a). However, SEC analysis of the H5N8-HA showed only a minor peak at 12.08 ml and a major peak at 14.73 ml from the Superose-6 column (Supplementary Fig. 1b).

Structure of the full-length H5N1-HA

Cryo-EM analysis of the SEC elution peak of the H5N1-HA at 14.25 ml showed an HA trimer structure that is similar to the reported structures[21]. The structures were determined at a resolution of 5.8 Å for the H5N1-HA. The extracellular domain densities agree well with the reported crystal structures of H5N1-HA (Supplementary Fig. 2a)[21]. The transmembrane helices of the HA are blurry (Supplementary Fig. 2a). Further cryo-EM analysis of the HA trimer in complex with the fragment antigen-binding (Fab) domain of the neutralization antibody H5M9 (FabH5M9)[22] improved the reconstruction, yielding a density map of the H5N1-HA and FabH5M9 complex at a resolution of 3.6 Å (Fig. 1 and Supplementary Figs. 3, 4, 5a). The structural model of the H5-HA complex was built with the crystal structure of the H5N1 HA-FabH5M9 complex (PDB: 4MHH) as a reference and was refined in real space. Residues 338–346 and 521–565 of the HA were not built due to the disordered density (Supplementary Table 1).

There are 6 predicted N-glycosylation sites in the H5N1-HA, including N21, N33, N158, N169, N289 in HA1 and N483 in HA2. Densities of the glycans could be observed at all 6 predicted glycosylation sites (Fig. 1). Mass spectrum analysis of the glycans at each site showed that 21% of the N-glycosylations have sialic acid modification (Supplementary Table 2). Lectin binding ELISAs showed that the complex glycans of the HA have both the α-2, 3 linked and the α-2, 6 linked sialic acid (Supplementary Fig. 5b). However, only the N-acetylglucosamine (NAG) and beta-D-mannose (BMA) in the core part of the oligosaccharide are visible in the density map due to the flexible nature and the heterogeneity of the glycans (Fig. 1 and Supplementary Fig. 5a).

The glycan-mediated interactions between the H5N1 HA trimers

Cryo-EM analysis of the SEC peak of the H5N1-HA at 12.56 ml showed long rod-shaped particles (Supplementary Fig. 2b). Three-dimensional classifications of the particles indicated that most of the particles are filaments with two HA trimers. Similarly, to further improve the reconstruction and obtain more details on the HA filaments, we used the FabH5M9 to improve particle alignments in the EM reconstructions (Supplementary Figs. 3, 4). Three-dimensional classifications showed that the particles could be grouped into three different major conformations A, B and C (Supplementary Fig. 3). After several rounds of classifications and refinements, the reconstructions were calculated at a resolution of 3.99 Å for conformation A, 6.19 Å for conformation B and 5.71 Å for conformation C (Supplementary Fig. 4). The filaments in conformation A are symmetric, while the filaments in conformations B and C are asymmetric (Fig. 2a). Structural models of conformations B and C were built by fitting the single HA-Fab trimer model into the corresponding density maps (Fig. 2a). The symmetric HA filament contains two head-to-head HA trimers, which are arranged in D3 symmetry. The HA1 heads face each other, and the transmembrane helices of the HA trimers are located at the distal ends of the filament (Fig. 2a). In the asymmetric HA filaments, the two HA trimers are not aligned along the three-fold axis, although the HA1 heads are also in proximity to each other. One HA trimer tilts for approximately 4.9° and 56.1° away from the three-fold axis of the other HA trimer in conformations B and C, respectively (Fig. 2a). Among the filamentous particles, particles in conformation A account for 25% of the total particles, while particles in conformations B and C account for 44% and 30% of the total particles, respectively (Supplementary Fig. 3).

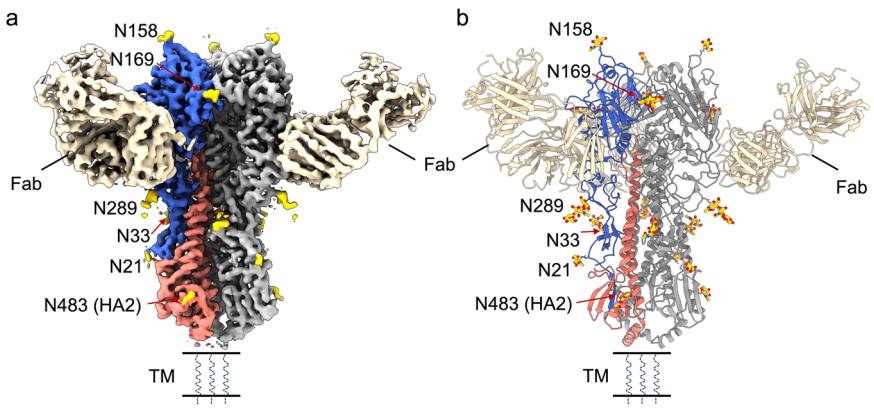
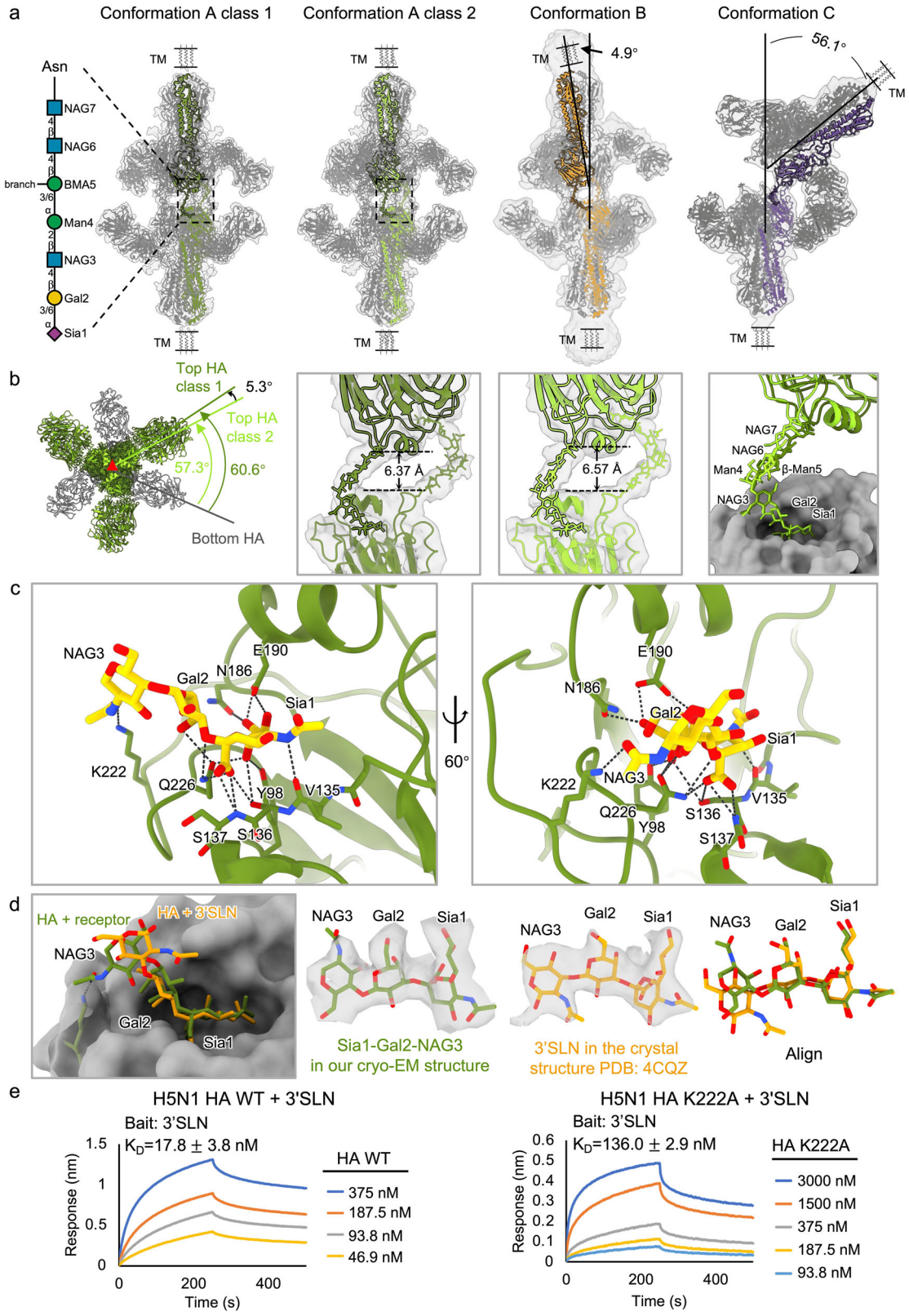


Fig. 1 | Structure of the full-length H5N1 HA trimer in complex with FabH5M9. a Cryo-EM density map of the H5N1 HA in complex with FabH5M9 shown in a surface-rendered representation. Densities of the glycans are colored yellow. The density map is set at a contouring level of 0.0194 $e^{-}/Å^{3}$. The flexible transmembrane domains are not resolved in the density map and are shown as a cartoon diagram. **b** Ribbon diagrams show the complex structure of the H5N1 HA and FabH5M9. The H5M9 Fabs are colored light yellow. One of the HA2 subunits is colored salmon, and one of the HA1 subunits is colored blue. The sugar moieties are shown in sticks with C, N and O atoms colored yellow, blue and red, respectively.



Structure of the symmetric H5N1-HA filament

Structural analysis of the symmetric HA filament showed that the two HA1 heads have no direct interactions (Fig. 2a). The interactions between the two HA trimers are mediated by the glycans attached to the residue N158, which is located at the top tip of the HA1 head (Fig. 2a). One HA trimer has a relative rotation of approximately 60.6° around the three-fold axis against the other HA trimer so that the six HA protomers have a zigzag arrangement, with the protruding glycans from the residues N158 directing to and binding the receptor binding grooves of the corresponding HA1s at the opposite side (Fig. 2a, b). The six glycan chains crosslinking the two trimers have a similar length of approximately 24.6 Å and are featured with a branch in the middle of the chain (Supplementary Fig. 5c). In the initial 3D classifications without

Fig. 2 | Structure of the filamentous H5N1 HA. a Different conformations of the filamentous H5N1 HA. The density maps are set transparent. The fitted HA-FabH5M9 structures are shown in ribbons. For the ribbon structures in conformation A, one pair of the HA protomers in each filament is colored green (dark green in class 1 and light green in class 2), while other HA protomers are colored gray. In the ribbon structure of conformation B, the HA protomers with the most significant conformation change, when compared with those in conformation A, are colored orange. In the ribbon structure of conformation C, the only pair of the HA protomers with N158 glycan chain-mediated interactions is colored purple. The sequence of the N158 glycan chain is shown as a cartoon diagram on the left. Blue squares, N-acetylglucosamine, NAG; green circles, mannose, Man; yellow circles, galactose, Gal; purple diamonds, sialic acid, Sia. **b** Structural comparisons of the two subclasses in conformation A. The vertical distances were measured between two parallel planes defined by N158s from the two HA trimers. **c** Detailed interactions between the glycan moiety and residues in the RBS of the HA. **d** Structural comparisons of the bound glycan moieties. 3' SLN is the receptor glycan analog that has the structure of Siaα2-3Galβ1-4NAGβ (PDB: 4CQZ). **e** BLI analyses of the binding of the receptor glycan analog 3' SLN to the wild-type H5N1 HA and the H5N1 HA mutant K222A. The K_D value presented is the average of two independent measurements.

symmetry imposed, densities of the six glycan chains are nearly identical (Supplementary Fig. 3), indicating that the six glycan chains have the same components and structure. Further classifications of the particles showed that the relative rotational positions between the two HA trimers vary in different classes (Fig. 2b). The variation in the rotation angle (from 50.3° to 64.9°) is correlated with the degree of twist in the glycan chains, indicating the flexible nature of the glycan chains (Supplementary Fig. 3, Supplementary Movie 1). The flexibility, however, is the key factor that obscures the high-resolution structure determination of the H5N1-HA filament. We, thus, split each head-to-head particle into two HA trimer particles and performed 3D classifications and refinements against the split HA trimers (Supplementary Fig. 3).

The reconstruction of the split HA trimers resulted in a density map at a resolution of 3.45 Å (Supplementary Figs. 3, 4, 5a). Densities of the sugar moieties at the proximate and distal ends are clearly visible in the final reconstruction (Supplementary Fig. 5a). However, densities of the sugar moieties located in the middle of the glycan chain are smeared, confirming the flexible nature of the sugar chains that mediate the head-to-head interactions of the two HA trimers. Based on the density map, we can build three sugar residues (NAG-Gal-Sia) at the distal end and two sugar residues (NAG-NAG) at the proximate end. By combining the mass spectrum analysis and oligosaccharide structures from the Glygen database[23], the core region of the glycan chain at N158 contains 7 monosaccharides, which from the distal end to the proximate end are Sia1-α2, 3-Gal2-β1, 4-NAG3-β1, 2-Man4-α1, 3/6-BMA5-β1, 4-NAG6-β1, 4-NAG7 (Supplementary Table 2). A branch extends from BMA5 (Supplementary Fig. 5c). Based on this information, we built the structure of the whole glycan chain into the density map of the symmetric H5N1-HA filament (Fig. 2a, b).

The glycan chain attached to N158 in each HA monomer extends and binds the receptor binding site (RBS) of the HA at the opposite side (Fig. 2b). There are totally three pairs of glycan-RBS interactions. Similar to previous studies, binding of the glycan to the receptor binding pocket of the H5N1-HA is mainly through the distal α-2, 3 linked sialic acid and galactose moiety (Sia1-Gal2) (Fig. 2b, c). The orientation and position of the distal Sia1-Gal2 moiety in the receptor binding pocket have no significant differences when compared with those of the reported crystal structures of the receptor analog and HA complexes (Fig. 2d)[24]. However, the third residue NAG3 at the distal end is in a completely different direction and location, which is likely attributed to the constraints introduced by the glycan-anchored proteins (Fig. 2d). Further structural analysis showed that the side chain of K222 forms a hydrogen bond with the acetylamino group of NAG3 (Fig. 2c), which has not been observed in previous studies. To explore the physiological function of this hydrogen bond, we immobilized the biotinylated avian receptor homolog 3'-Sialyl-N-acetyllactosamine (3' SLN) onto streptavidin biosensors to mimic the anchored receptor. The interaction of HA and 3' SLN was examined by using the biolayer interferometry (BLI) and mutagenesis studies. BLI analysis showed that 3' SLN binds H5N1-HA with a K_D of 17.8 nM (Fig. 2e). The mutation K222A significantly reduced the binding affinity of the H5N1-HA (K_D: 136.0 nM), indicating that the interaction between K222 and NAG3 plays a role in the specific recognition of the receptor.

Structures of the asymmetric H5N1-HA filaments

The asymmetric H5N1-HA filaments in conformation B have only two and a half pairs of the N158 glycan chain-mediated interactions. The lack of one N158 glycan-HA interaction reduces the restraints on the HA spikes, resulting in the tilt of one HA trimer away from the site where the glycan chain is missing. The tilt angle is approximately 4.9° (Fig. 2a). Comparisons of the glycan chains showed that the one in pairing with the missing glycan chain has a more extended conformation, whereas the glycan chains of the other two pairs adopt a more bent conformation when compared with these of the glycan chains from the symmetric filament (Fig. 3).

The asymmetric H5N1-HA filaments in conformation C have only one pair of the N158 glycan chain-mediated interactions. The HA trimer has a large tilt angle of 56.1° (Fig. 2a). The two visible glycan chains have similar conformation. Compared to these in the symmetric filaments and the asymmetric filaments in conformation B, the two glycan chains have the largest bending angle.

Structure comparisons of the glycan chains showed that the distal two sugar residues are fixed in the RBS, while positions of the third sugar residue vary in different glycan chains (Fig. 4a and Supplementary Fig. 5d). Except for residue K222, residue K193 could also be in close contact with the third sugar residue NAG3 (Fig. 4a). Based on structural modeling results, NAG3 can flip to establish a hydrogen bond with K193 (Fig. 4b). To confirm the possible interactions with K193, we, thus, mutated residue K193 to alanine. BLI analysis showed that K193A also significantly reduced the binding affinity of the H5N1-HA to 3' SLN (K_D: 100.2 nM) (Fig. 4c).

The glycan-mediated interaction within the protomers of the H5N8-HA

In December 2020, the first case of human infection with avian H5N8 was reported, despite the virus having circulated in poultry and wild birds since 2016[25]. Compared to the H5N1-HA, the HA protein of the H5N8 strain does not contain the N158 glycosylation site at the tip of the globular HA1 head (Supplementary Fig. 6). Instead, the H5N8 HA1 head contains the conserved N169 glycosylation site (Supplementary Fig. 6), which is approximately 30 Å away from the N158 site. It is unlikely that the H5N8 HA can mediate the formation of filament-like structures through head-to-head interactions. To further investigate the possible new modes of interactions between the glycans and the H5N8 HA, we prepared the H5N8 HA sample. SEC analysis of the H5N8 HA showed only a tiny peak at 12.08 ml but a dominant peak at 14.73 ml (Supplementary Fig. 1). We determined the cryo-EM structure of the H5N8 HA at a resolution of 2.8 Å using the sample prepared from the 14.73 ml peak (Supplementary Figs. 7, 8). Densities of the attached glycans could be observed on 5 glycosylation sites of the H5N8 HA, including N21, N33, N169, N289 and N483. Notably, the N169-associated glycan chain from one HA protomer binds the RBS of the neighboring protomer (Fig. 5a), indicating that the recombinant H5N8 HA is in an auto-inhibition state. We also performed cryo-EM analysis on the peak at 12.08 ml, and the results showed an asymmetric association of two HA trimers, in which the two HA trimers are aligned side by side (Supplementary Fig. 9). In comparison with the projections of the model, the interactions between the two HA trimers are mainly mediated by the glycans attached to N289 of HA1 (Supplementary Fig. 9).

Densities of the N169-associated glycan chain can accommodate 7 monosaccharides (Sia1-α2, 3-Gal2-β1, 4-NAG3-β1, 2-Man4-α1, 3/6-BMA5-β1, 4-NAG6-β1, 4-NAG7) (Fig. 5 and Supplementary Fig. 10). The

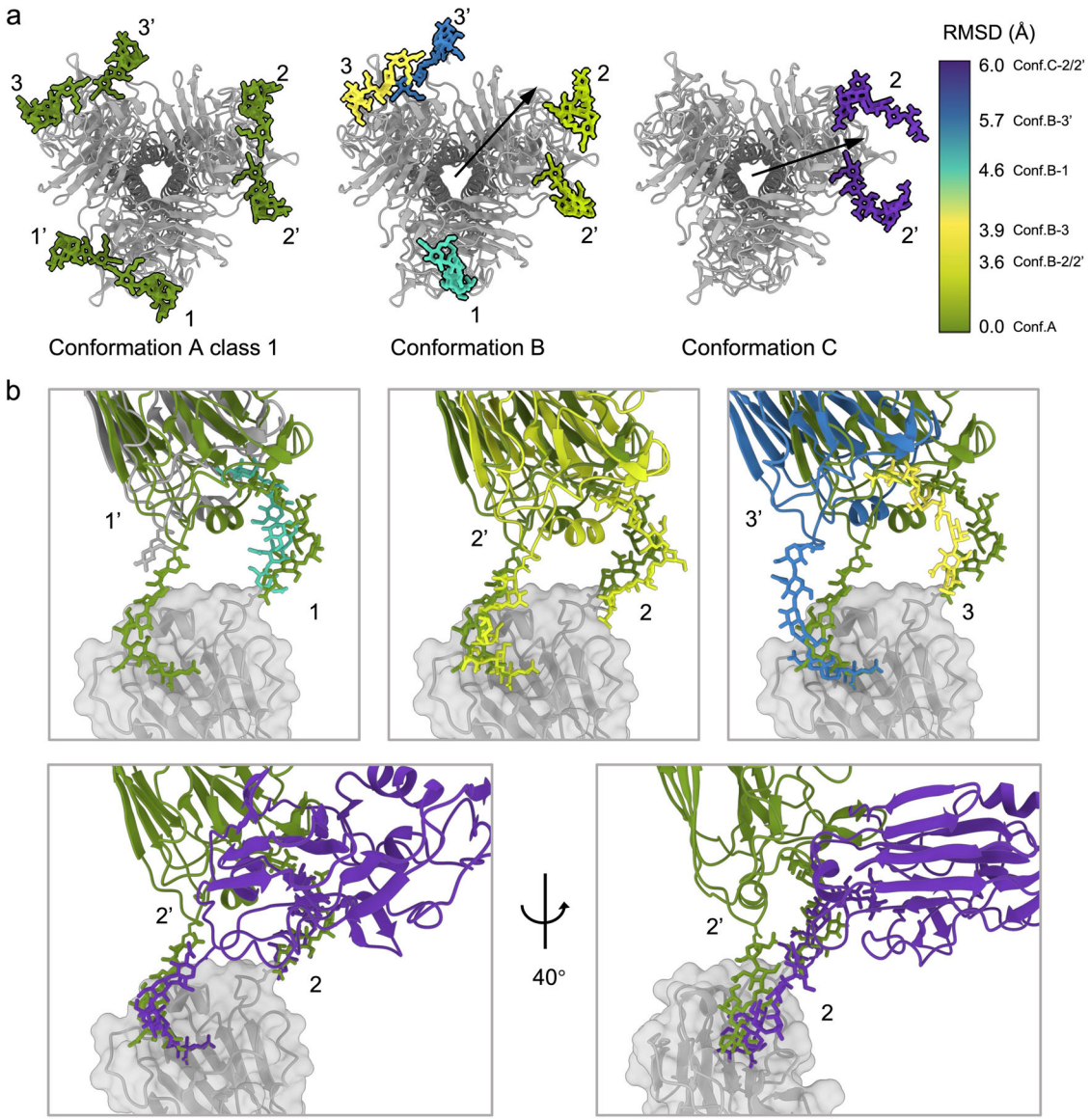


Fig. 3 | Structural comparisons of the glycan chains in different conformations of the filamentous HA. a Structures of the glycan chains in different conformations viewed from the top of the HA spike. The glycan chains are colored from green to purple, based on their root mean square deviations from the symmetric position as in conformation A. **b** Structural superimpositions of the glycan chains in different conformations. The models are superimposed based on the protomers in the HA spike at the bottom of the filament. The glycan chains and the corresponding HA protomer are colored from green to purple, based on their root mean square deviations from the symmetric position as in conformation A.

glycan chain extends from N169 but turns sharply in the middle toward the receptor binding site of the nearby HA protomer. Residues constituting the receptor binding sites of the H5N8 and H5N1 HAs are highly conserved (Fig. 5b). Differences were identified at 3 positions around the receptor binding site, including positions 137, 222 and 227. These positions of the H5N1 HA are occupied by S137, K222 and S227, respectively, whereas in the H5N8 HA, these are occupied by residues A137, Q222 and R227, respectively (Fig. 5b). The Sia1-Gal2 moiety forms hydrogen bonds with Y98, V135, S136, A137, H183, E190 and Q226 in the RBS pocket, which are similar to these in other reported receptor-HA interactions (Fig. 5b and Supplementary Fig. 10c). However, compared with these binding the H5N1 HA, the sugar moiety in the RBS of the H5N8 HA has a rotation of approximately 16° toward the peptide fragment 135–138 with the acetyl group of Sia1 as the pivot point. Since the hydrogen bond formed between the main chain of V135 and the acetylamino group of Sia-1 is well maintained, the shift brought by the rotation pushes the peptide fragment 135–138 outward by 1.4 Å (Fig. 5c). The α-2, 3-bonded Gal2 forms two hydrogen bonds with Q226 (Fig. 5d). NAG3 rotates for approximately 60° when compared with that of the H5N1 HA (Fig. 5c). The acetyl group of NAG3 is in close proximity to residues Q222 and R227 (Fig. 5d). To verify the possible interaction with Q222 and R227, we further mutated Q222 and R227 to alanine. BLI analysis showed that neither the mutation Q222A nor R227A significantly affected the binding affinity with 3' SLN (WT-HA K_D: 17.7 nM, R227A-HA K_D: 28.1 nM) (Fig. 5e), indicating that unlike K222 of the H5N1 HA, Q222 or R227 of the H5N8 HA doesn't play a vital role in binding the receptor. Previous studies showed that the mutations K222Q and S227R in H5N8-HA could enable the binding of H5N8 HA to sialyl Lewis x[26,27]. Mass spectrometry analysis confirmed the presence of fucosylation in H5N8 HA. However, no additional fucose density was observed near NAG3 in the map likely due to its high flexibility or low occupancy.

Discussion

Sialic acid modification on glycoprotein or glycolipids has been certainly identified as the receptor of the influenza virus. Glycan arrays immobilized

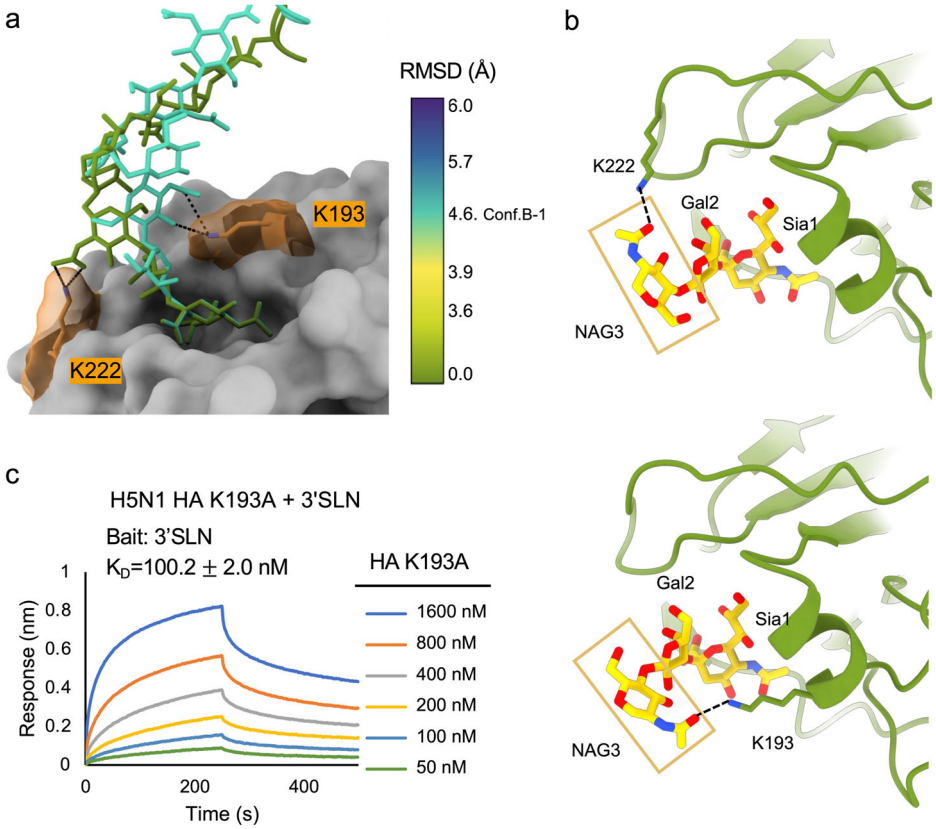


Fig. 4 | Interactions between the glycan and H5N1 HA. a Structural superimpositions of the glycans in class 1 of conformation A and the glycans at position 1 of conformation B revealing possible residues in contact with NAG3. **b** Ribbon and stick diagrams showing possible interactions between the glycan with K222 or K193. **c** BLI analysis of the interaction of 3'SLN with K193A HA. The K_D value represented is the average of three independent measurements.

with glycans in various structures are frequently used to investigate the binding specificity of different HAs. However, the glycans decorated on the host cell surface are complex. Until now, the exact intricate interaction between the receptor and the HA is unknown, and the structure of a protein-anchored glycan in complex with HA has not been reported.

HA proteins produced in mammalian cells and virions of some subtype influenza viruses tend to aggregate, and this aggregation is related to the glycosylation of HA[28]. Since the complex N-glycans produced in mammalian cells prevent the crystallization of HAs, many previous structural studies of HA employ incompletely glycosylated HAs, such as those produced in insect cells[21,29,30]. The complex N-glycans produced in mammalian cells contain terminal sialic acids, while the N-glycans produced in insect cells are simpler and mainly contain terminal mannose residues. We used the mammalian cells for HA production, which can reflect the authentic glycosylation pattern of the influenza virus receptor. The glycan-mediated interactions within the filamentous HAs may mimic the multivalent interactions between HA and receptors. It was reported that individual HA–Sia interaction is weak and dynamic[31]. Theoretically, multivalent binding modes, particularly symmetric ones akin to that in the conformation A of the filamentous HA, could exponentially enhance binding affinity compared to the monovalent mode[18,32]. The high binding affinity could allow avian IAVs to specifically recognize the few available receptors of the host cells in the human upper respiratory tract[14]. Evidence supporting increased binding affinity through multivalent interactions has been observed in HA-glycan interactions studied using arrays decorated with branched glycans[21,33]. Even for the same scaffold, the complex nature of the glycans allows different multivalent interactions, as displayed in the three conformations of the filamentous HA. A certain glycoprotein receptor in a homotrimer state has not been reported. As our study mainly captured conformational states with more than two pairs of HA-glycan interactions, we reason that glycosylated cell surface proteins in a certain oligomeric state would be the preferred carrier for the glycan receptor of the influenza virus. This finding may help to narrow down the search region for molecules that carry the glycan receptor.

NA cleaves cellular sialic acid residues and enables the release of newly assembled virions. We expressed NA from *Clostridium perfringens* and tested whether the observed filamentous HA can be destroyed in vitro. The multivalent interaction can be disrupted under 37 °C with the ratio of 1:5 (NA:HA, by weight) as shown by SEC and band shift on SDS-PAGE gels (Supplementary Fig. 11a). The SEC elution pattern of the digested sample showed a significant decay of the oligomer HA peak compared to the control without adding NA (Supplementary Fig. 11b). Sample of the NA-treated HA also displays a significant reduction of the filamentous dimers as shown in the analysis of cryoEM microscopy (Supplementary Fig. 11c). Thus, we postulate that the observed HA and receptor interaction form will not impede virus infection or release in vivo.

It was reported that the NAs of most H5N1 strains circulating in 2003–2011 have a short membrane proximate stalk structure, which results in lower enzyme activity when compared with those with a long stalk structure. The lower enzyme activity of NA causes incomplete cleavage of the distal sialic acid and self-aggregation of H5N1 virions[28,34]. The residual Sia on some of the HAs may help aggregate the H5N1 virions through the strong 'triple-pair' interactions.

H5N1-HA keeps the ability to bind to avian receptors with the α-2, 3 linked sialic acid. Sequence alignments of HAs reveal that the mutation of T160 to A160 causes the lack of N158 glycosylation in H5N8-HA (Supplementary Fig. 6). This loss is characteristic of more recent and currently circulating H5 subtype viruses. However, the N169 glycosylation site remains conserved in these H5 subtype viruses (from the GISAID database, https://www.gisaid.org/). The T160A mutation in HA has been reported to enhance the avidity to α-2, 6 linked sialic acid, which is the receptor of human influenza virus[35,36]. In the ferret transmissible variants, the HA protein also lacks the N158 glycosylation site[15,16]. The loss of N158 glycosylation would eliminate the strong 'triple-pair' interactions of the H5N1-HA and expose the site for binding the α-2, 6 linked sialic acid.

The interactions with NAG3, as observed in the filamentous H5N1 HA, indicate that additional interactions with the glycans could occur and are probably virus-strain dependent. Sequence alignments of the HAs from

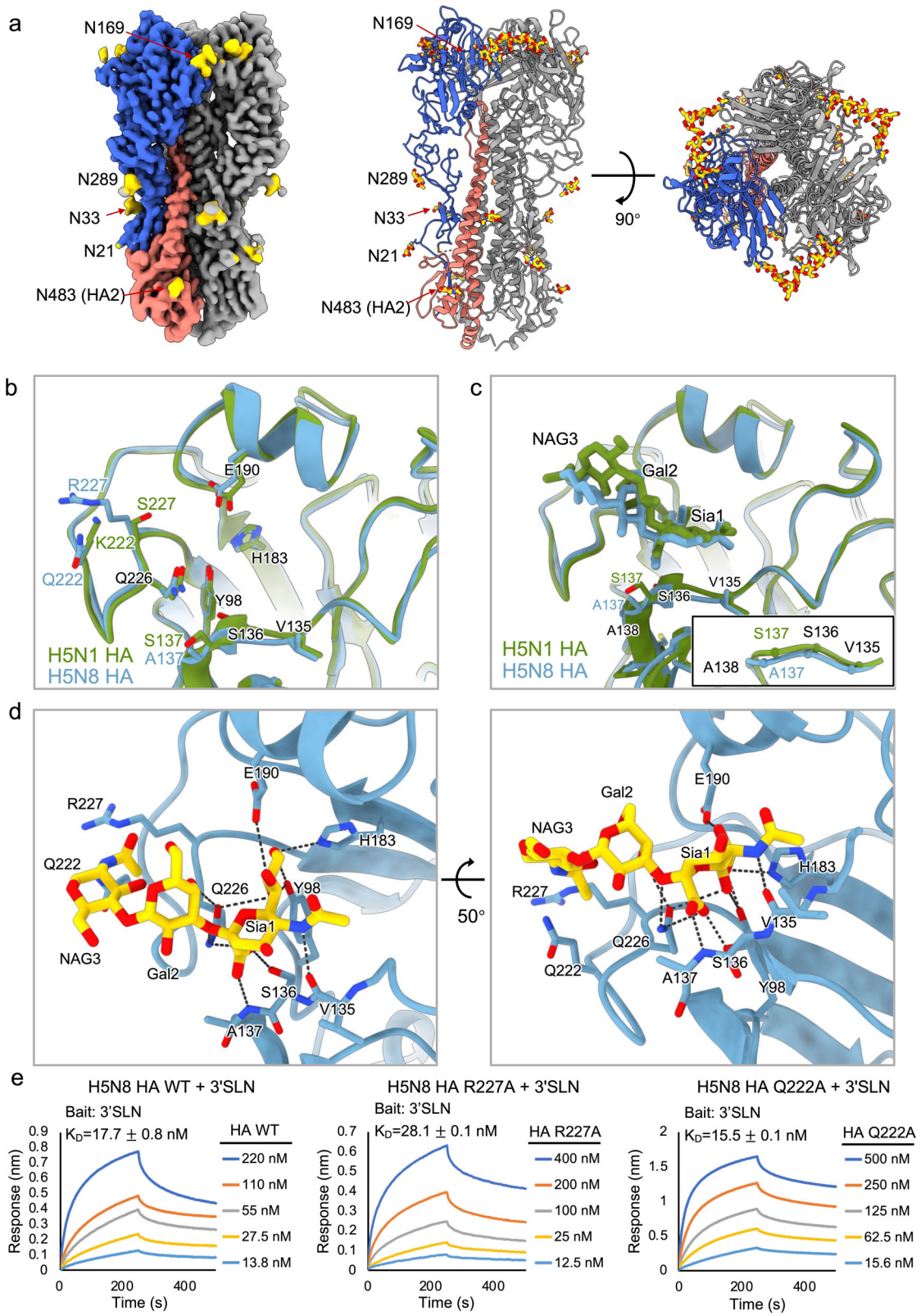


Fig. 5 | Cryo-EM structure of the H5N8 HA. a Left: A cryo-EM map of the H5N8 HA reconstruction shown in the surface-rendered representation. Densities of the sugar moieties are colored yellow. Middle and Right: ribbon diagrams showing the structure of the H5N8 HA in side and top views. The sugar moieties are shown in sticks with N, O and C atoms colored blue, red and yellow, respectively. The HA1 and HA2 of one protomer are colored blue and pink, respectively. The HA1s and HA2s in the other two protomers are colored gray. **b** Structural comparisons of the receptor binding regions of the H5N1 and H5N8 HAs. Residues in close proximity to the bound sugar moiety are shown in sticks with N and O atoms colored blue and red, respectively. The C atoms are colored according to the color of the corresponding ribbon. **c** Structural comparisons of the bound sugar moieties of the H5N1 and H5N8 HAs showing different conformations of the bound NAG3 and the 135–138 loop of HA. The sugar moieties are shown in sticks. **d** Detailed interactions between the bound sugar moiety and the H5N8 HA. **e** BLI analysis of the interactions of 3'SLN with WT HA and HA variants containing mutations in residues near NAG3 of the bound sugar moiety. The K_D values represented are the average of two independent measurements.

different IAV strains showed that residues 193 and 222 of HA1 are not conserved among different virus strains (Supplementary Fig. 6). Previous studies showed that residues 193 and 222 can interact with and help to fix the α-2, 6 Sia-Gal moiety of the human receptor[37–39]. Residue K222 of the H1N1-HA can directly interact with the Gal2 of the human receptor through a hydrogen bond (Supplementary Fig. 12). Residue S193 of the H1N1-HA is in close proximity to the sugar residue Gal4 of the human receptor (Supplementary Fig. 12)[38]. These results indicate that residues K193 and K222 can also interact with the α-2, 3 linked glycan chain. Given the variation at these two positions, residues 193 and 222 could render certain virus strains a higher binding affinity to a certain receptor or even affect the host tropism of the virus[27,40–42]. Structural comparisons with other HA-glycan interactions show that positions of the bound Sia1s are all similar, and residues around the bound Sia1 are highly conserved[30,43–45] (Supplementary Fig. 13). These conserved residues include Y98, W153, H183, and L/I194. However, the positions of the bound Gal2 and NAG3 vary. Correspondingly, residues around the bound Gal2 and NAG3 vary. These varied positions mainly involve residues 136, 186, 190, 193, 222, and 226 (Supplementary Fig. 13). The variations in position and residue types could be determinants for receptor preference of different HAs. Observations in previous studies and this study imply that the RBD of HA keeps the Sia1 binding site conversed but varies the binding sites for other glycan residues to establish complex interactions with the glycan receptors.

It is reported that monosaccharide-modified HA protein could induce a stronger and cross-strain protective effect than fully glycosylated HA in mice immunization[46]. The glycosylation of influenza vaccines can influence the immunogenicity. The RBS region of the mammalian cell expressed HA could be occupied by terminal sialic acid of its own N-glycosylation modifications. Recent recombinant HA subunit vaccines are mainly produced by eukaryotic cells, including mammalian or insect expression systems. The recombinant HA protein produced from the mammalian system carries complex N-glycosylation, and the essential antigenic region RBS could be occupied by sialic acid, which may affect the immunogenicity of recombinant HA vaccines.

Surfactant protein in the upper respiratory tract can block the infection of flu virus[47,48]. The carbohydrate recognition domain (CRD) of surfactant protein-D (SP-D) binds to mannose-rich glycans on HA and NA with the help of Ca^{2+}, which is an important innate immune barrier to influenza virus infection[49–52]. The glycosylation site N165 of the H3-HA is highly glycosylated with mannose-rich glycans, and the binding of SP-D could interfere with the receptor binding at the RBS[53,54]. Here we show that different from the H3-HA, the N169 site of H5N8 HA around the RBS is of complex glycans that contain α-2, 3 linked terminal sialic acids, which is consistent with previous studies[55]. Thus, the H5-HAs should be resistant to SP-D binding and could escape the block mediated by SP-D.

Represented by H5N8 HA in this study, currently spreading H5 subtype HAs possess both N158 glycosylation deletion and conserved N169 glycosylation site. The auto-inhibition state of such HAs could destroy the epitope around the RBS, which is a key epitope recognized by neutralization antibodies. Thus, special consideration should be made for the development of recombinant subunit vaccines against the H5N8 IAV. According to GISAID statistics, the N158 glycosylation site is dominant in HAs of H5Nx avian influenza virus circulating around 2004 and in H3N2 HAs from 2013 to present[19,56]. The N158 glycosylation of these recombinant HA vaccines could cause the dimerization of two HA trimers and impair epitopes important for the induction of neutralization antibodies.

Materials and methods

Gene synthesis and cloning

The genes coding H5N1-HA (A/Vietnam/1203/2004, Group 1) (Uniprot ID: Q6DQ33), H5N1-HA head antibody H5M9 (PDB ID: 4MHH), H5N8-HA and NA (A/chicken/Czech_Republic/1566-1/2021) were synthesized by Qinglan Biotech co., Wuxi, China. The codons of the genes were optimized for mammalian expression. The H5N1-NA gene (Uniprot ID: P10481.1) was optimized for *E. coli* expression and was also synthesized by Qinglan Biotech Co., Wuxi, China. Genes of the full-length H5N1-HA and the ectodomain of H5N8-HA (1-526) were PCR amplified and were then cloned into the vector pCMV, which introduces a C-terminal twin-strep tag to the recombinant proteins. Genes of the antibody heavy and light chains were cloned into a specialized vector with fragments that encode a signal peptide and the human antibody constant region. The gene encoding the ectodomain of the H5N1-NA was cloned into the vector pET22b that introduces an N-terminus 6×His tag to the recombinant protein. For BLI analysis, the ectodomains of H5N1-HA and H5N8-HA, along with the T4 trimerization peptide and a flag tag, were cloned into the vector pCMV. The gene encoding the full-length H5N8-NA was cloned into the vector pCMV that introduces a C-terminus 6×His tag to the recombinant protein.

Expression and purification of the H5N1-HA

The plasmids were amplified and purified using standard plasmid extraction kits (Tiangen, Inc., Beijing, China). HEK293F cells were grown in SMM 293-TI medium (Sino Biological Inc.) supplemented with 0.5% (v/v) FBS. For one liter cell culture, 2 mg plasmids were pre-incubated with 8 mg polyethylenimine (PEI) for 20 min before being used for cell transfection. The cells were transiently transfected when the cell density reached 2.5–3 × 10^6/ml. Cells were harvested 48 h post-transfection by centrifugation at 1000 × *g*. Cell pellet was resuspended in a lysis buffer containing 20 mM HEPES at pH 8.0, 150 mM NaCl and protease inhibitor cocktail. The resuspended cells were sonicated, and the pellet of cell membrane was collected by centrifugation at 100,000 × *g* for 1 h at 4 °C. The membrane was resuspended in the lysis buffer containing 1% (w/v) Lauryl Maltose Neopentyl Glycol (LMNG, Anatrace). After the removal of the insoluble pellet by centrifugation at 150,000 × *g* for 30 min, the supernatant was collected and applied to Strep-Tactin beads (IBA). The beads were washed with 3 column volume of the washing buffer (20 mM HEPES pH 8.0, 150 mM NaCl, 0.003% LMNG) and eluted with the washing buffer containing 5 mM D-desthiobiotin (Sigma). Concentrated elution fractions were further purified through size exclusion chromatography with a Superose-6 Increase column (Cytiva). Fractions of the peaks were collected and concentrated to approximately 0.3 mg/ml for cryo-EM sample preparation.

Expression, production and purification of FabH5M9

The plasmids were prepared following the same procedure as for the H5N1-HA plasmid. For transfection, one liter of cells was transfected with 1.5 mg of heavy chain plasmid, 1.5 mg of light chain plasmid, and 12 mg of PEI. The supernatant was collected 72 h post-transfection. Following concentration and buffer exchange, the medium was loaded to Protein A Sepharose beads (Cytiva). The beads were washed with a buffer containing 20 mM HEPES pH 8.0 and 150 mM NaCl. Subsequently, the antibody was eluted by using 0.1 M glycine at pH 3.5. The elution was immediately neutralized by 1 M Tris-HCl at pH 9.0.

To obtain the Fab fragment of H5M9, the protease HRV3C with a GST tag was utilized to cleave the Fc part. The enzyme was removed via affinity purification using Glutathione Sepharose 4B beads (Cytiva). The Fc portion was then eliminated through affinity purification using Protein A Sepharose beads (Cytiva).

Preparation of the H5N1-HA and FabH5M9 complex

The purified H5N1-HA and FabH5M9 were mixed at a molar ratio of 1:3 (HA:Fab). After incubation at 4 °C overnight, the complex was further purified by size exclusion chromatography using a Superose-6 Increase column (Cytiva) running in a buffer containing 20 mM HEPES at pH 8.0, 150 mM NaCl and 0.003% (w/v) LMNG. Fractions of the HA-filament peak were collected and concentrated to approximately 0.3 mg/ml for cryo-EM sample preparation.

Expression and purification of the H5N8-HA ectodomain

For 1 L cell culture, the cells at a density of 2.5–3 × 10^6/ml were transfected with 2 mg plasmid and 8 mg PEI. The supernatant of the cell culture was collected 72 h post-transfection and was then buffer changed to 20 mM

HEPES at pH 8.0, 150 mM NaCl. The supernatant was loaded onto Strep-Tactin beads. The beads were washed with the buffer containing 20 mM HEPES at pH 7.8 and 150 mM NaCl. The recombinant HA ectodomain was then eluted with the same buffer supplemented with 5 mM D-desthiobiotin.

Detecting the sialic acid modification by ELISA

For this, 100 ng/well HA was coated on ELISA plates (Costar) at 4 °C overnight. The plate was blocked by 100 µl carbo-free blocking solution (Vector) for 1 h at room temperature. Then the plate was incubated with 100 µl biotinylated SNA (Vector, 1:2000 diluted in carbo-free blocking solution) or MAL II (Vector, 1:200 diluted in carbo-free blocking solution) for 1 h. Each well was washed with PBS containing 0.1% (v/v) Tween 20 (PBST) six times. The horseradish peroxidase (HRP) conjugated streptavidin (Proteintech) was diluted 1:1000 in carbo-free blocking solution and incubated for 30 min (100 µl/well) followed by PBST washing. 100 µl 3,3',5,5'-tetramethylbenzidine (TMB) solution (Biopanda) was added to each well and incubated for 5 min. The reaction was stopped by adding 50 µl 1 M H_2SO_4. The signal was recorded by spectrophotometer (BioTek) at OD450 nm.

N-glycosylation of HA analyzed by mass spectrometry

The SEC eluted samples of the H5N1 HA and H5N8 HA were analyzed by SDS-PAGE gels. The gel bands of HA0, HA1 and HA2 were excised from the gel, reduced with 5 mM of DTT and alkylated with 11 mM iodoacetamide, which was followed by in-gel digestion with sequencing grade modified trypsin at 37 °C overnight. The peptides were extracted twice with 0.1% (v/v) trifluoroacetic acid in 50% (v/v) acetonitrile aqueous solution for 30 min and then dried in a SpeedVac (Thermo Fisher). Peptides were redissolved in 20 µl 0.1% (v/v) trifluoroacetic acid and 6 µl of extracted peptides were analyzed by Thermo Scientific Q Exactive HFX mass spectrometer. For liquid chromatography-tandem mass spectrometry (LC-MS/MS) analysis, the peptides were separated by a 120 min gradient elution at a flow rate of 0.30 µl/min with a Thermo-Dionex Ultimate 3000 HPLC system, which was directly interfaced with a Thermo Scientific Q Exactive HFX mass spectrometer. The analytical column was a homemade fused silica capillary column (75 µm ID, 350 mm length) packed with C-18 resin (1.9 µm, Dr. Maisch GmbH). The mobile phase consisted of 0.1% (v/v) formic acid, and mobile phase B consisted of 80% (v/v) acetonitrile and 0.1% (v/v) formic acid. Q Exactive HFX mass spectrometer was operated in the data-dependent acquisition mode using Xcalibur 4.5.445.18 software, and there was a single full-scan mass spectrum in the orbitrap (300–1800 m/z, 60,000 resolution). The normalized higher-energy collision disassociation (HCD) fragmentation energy steps were set to 27%, 30% and 33%.

The tandem mass spectrometry (MS/MS) spectra from each LC-MS/MS run were searched against the influenza virus database using PMi-Byonic (Version 2.11.0). The search criteria were as follows: trypsin was chosen as the specific enzyme; two missed cleavages were allowed; carbamidomethylation (C) was set as fixed modification; the oxidation (M) was set as the variable modification; precursor ion mass tolerance was set at 20 ppm for all MS acquired in an orbitrap mass analyzer; and the fragment ion mass tolerance was set at 0.02 Da for all MS2 spectra. Confidence levels were set to 1% FDR (high confidence). The MS data was searched from the mammalian N-glycan library. According to the Byonic score, results with scores lower than 30 were removed.

According to the glycan composition determined by Byonic, we searched for the corresponding glycan structure from the Glygen database by the composition search functionality. We selected N-glycosylation results from human samples containing α-2, 3-linked sialic acids for presentation (Supplementary Table 2).

NA expression and NA cleavage assay

The NA-pET22b plasmid was transformed into *Escherichia coli* Rosetta cells (Novagen) for the production of recombinant NA. Cultivation of the cells was carried out at 37 °C until reaching an OD600 value of approximately 0.6. Subsequently, the cultivation temperature was lowered to 16 °C, and Isopropyl β-D-1-thiogalactopyranoside (IPTG) was added to a final concentration of 0.5 mM to induce expression of the recombinant protein for 17 h. Cells were harvested by centrifugation at 4000 rpm and lysed by sonication using a lysis buffer containing 20 mM HEPES at pH 7.5 and 150 mM NaCl. The cell membrane pellet was collected by ultra-centrifugation at 100,000×*g* for 1 h at 4 °C.

The membrane was then resuspended in the lysis buffer containing 2% (w/v) Dodecyl Maltoside (DDM, Anatrace) and incubated at 4 °C for 2 h. Following ultra-centrifugation at 150,000×*g* for 30 min, the supernatant was collected and applied to cobalt-charged resins (BD TALONTM). The resin was washed by the lysis buffer containing 0.02% (w/v) DDM and 20 mM imidazole. Recombinant NA was eluted from cobalt resin using the lysis buffer containing 0.02% (w/v) DDM and 300 mM imidazole. The collected elution fractions were concentrated and further purified by SEC using a Superdex 200 increase (Cytiva) column with a buffer containing 20 mM HEPES at pH 7.5, 150 mM NaCl and 0.02% (w/v) DDM.

As NA monomer is of lower activity, we only selected NA oligomer fractions for the following HA glycan cleavage assays. Briefly, NA was added into HA elutes with a ratio of 1:5 (NA : HA, w/w), and the reaction mixture was incubated at 37 °C for 30 min, 1 h, 2 h or 4 h, respectively. The cleavage efficiency was analyzed by using SDS-PAGE gels. Compared to the control group without NA, a treatment of 4 h showed an obvious band shift.

Cryo-EM sample preparation and data collection

Three-microliter aliquots of purified full-length H5N1-HA or the H5N1-HA and 4MHH_H5M9 complex, at a concentration of 0.3 mg/ml, were applied to glow-discharged Lacey carbon grids (TED PELLA, Cu 400 mesh). To address the preferred orientation problem[57], the ectodomain of H5N8-HA was mixed with CTAB (final concentration 4 mM) before being loaded onto glow-discharged holey carbon grids (Quantifoil, Cu 400 mesh, R1.2/1.3). Subsequently, the grids were blotted and plunged into liquid ethane using a Vitrobot Mark IV (Thermo Fisher).

Images of the H5N1-HA and the H5N1-HA-Fab complex were recorded using a Titan Krios electron microscope (Thermo Fisher) operating at an acceleration voltage of 300 kV, equipped with a Gatan K2 Summit Camera. The defocus series ranged from −2 µm to −3 µm. The pixel size was 1.32 Å, and each stack was exposed for 8 s with a total dose of approximately 50 electrons per $Å^2$. Images of the H5N8-HA ectodomain were collected using a Titan Krios electron microscope (Thermo Fisher) operating at an acceleration voltage of 300 kV, equipped with a GIF Quantum energy filter (slit width 20 eV) and a Gatan K2 Summit camera. The defocus range was from −1.5 µm to −2.5 µm, with a pixel size of 1.0742 Å. Each image was dose-fractionated into 32 movie frames with a total exposure time of 8 s and a total dose of approximately 50 electrons per $Å^2$. Data collection was performed using SerialEM and AutoEMation2[58,59].

H5N1-HA-Fab cryo-EM data collection and processing

A total of 6096 movie stacks of the H5N1-HA-Fab complex were collected (Supplementary Table 1). The frames within each movie stack were aligned, summed, and 2× binned using MotionCor2 v1.2.6[60]. The Contrast Transfer Function (CTF) parameters of the micrographs were determined by Gctf v1.18[61]. Subsequently, a total of 1,197,104 particles were picked using Gautomatch v0.56.

2D classifications performed with RELION v3.1[62] yielded two major classes: one containing a single HA trimer, and the other with two HA trimers arranged in a filament-like structure. The particles from each class were split for independent 3D classifications in RELION v3.1 without imposing symmetry.

For the single HA trimer class, 806,267 particles were used for 3D classifications, and 198,231 particles were subjected to the final 3D refinement with C3 symmetry imposed, which resulted in a map at a resolution of 3.62 Å.

HA filament particles were classified into three major conformations (A, B, and C) by 3D classifications. Of these, 25% exhibited a head-to-head symmetric conformation (conformation A). Subsequent 3D classifications

further divided these symmetric particles into two dominant classes, differing in relative rotations between the head-to-head HA trimers (approximately 60° in class A1 and approximately 57° in class A2). For head-to-head HAs in class A1, 38,100 particles were selected for cryoSPARC v4.4.1[63] non-uniform refinement with D3 symmetry imposed, resulting in a map at a resolution of 3.99 Å. For head-to-head HAs in class A2, 46,732 particles were selected for cryoSPARC non-uniform refinement with D3 symmetry imposed, resulting in a map at a resolution of 4.25 Å (Supplementary Figs. 3, 4 and Supplementary Table 1). To analyze the flexibility of head-to-head HA trimers, symmetric filamentous particles were classified by cryoSPARC 3D classification without symmetry imposed. Five classes of symmetric HA filaments exhibiting different relative rotation angles were obtained. Further 3D variability analysis indicated that the relative rotation angle of symmetric HA filaments varied from 50.3° to 64.9°. The twisting trajectory of the HA filament was recorded as a movie using Chimera[64] (Supplementary Fig. 3, Supplementary Movie 1).

The filaments in conformations B and C are asymmetric. Approximately 44% of HA filament particles have one HA trimer tilting slightly relative to the other HA trimer within the filament (Conformation B). A total of 60,487 particles exhibiting this conformation were selected for cryoSPARC non-uniform refinement without symmetry imposed, resulting in a map with a resolution of 6.19 Å. Furthermore, approximately 30% of HA filament particles have one HA trimer tilting significantly relative to the other HA trimer within the filament (Conformation C). A total of 86,297 particles exhibiting this conformation were selected for cryoSPARC non-uniform refinement without symmetry imposed, resulting in a map at a resolution of 5.71 Å (Supplementary Figs. 3, 4 and Supplementary Table 1).

To improve the reconstruction of the H5N1-HA symmetric filament, we split each filament into two HA trimer particles and performed 3D classifications and cryoSPARC non-uniform refinements against the split HA trimer particles. The final refinement resulted in an improved map with a resolution of 3.45 Å (Supplementary Figs. 3, 4).

H5N8-HA cryo-EM data collection and processing

A total of 4057 movie stacks of the H5N8-HA were collected, and the processing procedure was the same as that for the H5N1-HA-Fab dataset. A total of 2,728,124 particles were picked by Gautomatch v0.56 and subjected to 2D classifications. Subsequently, 1,409,642 particles with clear secondary structure features were selected for several rounds of 3D classifications without imposing symmetry. Particles in 2 classes with clear structural details were selected for 3D refinements with C3 symmetry imposed. To further remove the reconstructions, particles were subjected to 3D classifications without alignment and a total of 299,040 particles were then selected for cryoSPARC non-uniform refinements with C3 symmetry imposed. The final density map was applied with a negative B-factor of 126.1 $Å^2$, and the final resolution of the map is 2.8 Å (Supplementary Figs. 7, 8). The local resolution maps were calculated by cryoSPARC v4.4.1 with a threshold of 0.143.

Model building and structure refinement

Atomic models of the H5N1-HA-Fab, the split H5N1-HA-Fab from the HA filaments and the H5N8-HA were built with the H5N1-HA crystal structure as references (PDB: 2FK0)[21]. The models were adjusted by using COOT v0.9.6[65] and were refined by using the PHENIX v1.19[66] cryo-EM real-space refinement tool. Models of the low-resolution H5N1-HA filament reconstructions were built based on fitting the high-resolution structures into the density maps with Chimera[64]. The interactions between HA and receptor were analyzed by using PISA[67]. Detailed parameters and statistics for data collection and processing are shown in Supplementary Table 1. All structural representations were prepared by using UCSF ChimeraX v1.3[68].

Purification of the HA ectodomains for BLI analysis

The cell transfection procedure was the same as described above. The HA ectodomain and full-length NA were transfected separately. The cell cultures were then mixed with a volume ratio of 2:1 (HA:NA) 72 h post-transfection, and the mixed cells were co-cultured for 24 h to cleave the sialic acid of the HA. The supernatant was harvested and loaded to anti-Flag resin (Genescript). The resin was washed with a buffer containing 20 mM HEPES at pH 7.6 and 150 mM NaCl. HA was eluted by 3×flag peptide and concentrated for SEC analysis. We chose the later peak that contains the single HA trimer for BLI analysis.

Biolayer interferometry analysis

BLI analyses were conducted at 25 °C using a ForteBio Octet Red biosensor system (ForteBio). Biotinylated 3' SLN (GlycoNZ, 0036-BP) was immobilized onto Streptavidin biosensors (ForteBio, 18-5019) to achieve a signal of 1 nm. The sensors were washed in buffer containing 20 mM HEPES at pH 7.6, 150 mM NaCl and 0.05% (v/v) Tween 20 for 180 s to establish a baseline. Subsequently, the biosensors were immersed into wells containing various concentrations of HA and HA mutants for 250 s, followed by a 250 s dissociation step in the same buffer as the baseline step. BLI data were analyzed using Octet software (version 9.0, ForteBio) in the standard 1:1 binding mode.

Data availability

The atomic coordinates and EM maps have been deposited into the Protein Data Bank (http://www.pdb.org) and the EM Data Bank (http://www.emdataresource.org), respectively, with the accession numbers EMD-60016 (single H5N1 HA in complex with FabH5M9), EMD-60010 (conformation A class 1 of symmetric H5N1 HA filament), EMD-60011 (conformation A class 2 of symmetric H5N1 HA filament), EMD-60015 (H5N1 HA split from symmetric H5N1 HA filament), EMD-60012 (conformation B of asymmetric H5N1 HA filament), EMD-60013 (conformation C of asymmetric H5N1 HA filament), EMD-60014 (H5N8 HA bound with N169 sialylated glycan chain), 8ZDW (atomic models of H5N1 HA bound with Sia-Gal-NAG), 8ZDV (atomic models of H5N8 HA bound with Sia-Gal-NAG-Man-BMA-NAG-NAG).

Received: 11 June 2024; Accepted: 11 September 2024;
Published online: 17 October 2024

References

1. McGeoch, D., Fellner, P. & Newton, C. Influenza virus genome consists of eight distinct RNA species. *Proc. Natl Acad. Sci. USA* **73**, 3045–3049 (1976).
2. Yoon, S. W., Webby, R. J. & Webster, R. G. Evolution and ecology of influenza A viruses. *Curr. Top. Microbiol. Immunol.* **385**, 359–375 (2014).
3. Taubenberger, J. K. & Morens, D. M. The pathology of influenza virus infections. *Annu. Rev. Pathol.* **3**, 499–522 (2008).
4. Webster, R. G., Yakhno, M., Hinshaw, V. S., Bean, W. J. & Murti, K. G. Intestinal influenza: replication and characterization of influenza viruses in ducks. *Virology* **84**, 268–278 (1978).
5. Skehel, J. An overview of influenza haemagglutinin and neuraminidase. *Biologicals* **37**, 177–178 (2009).
6. Cohen, M. et al. Influenza A penetrates host mucus by cleaving sialic acids with neuraminidase. *Virol. J.* **10**, 321 (2013).
7. Chen, F. et al. Key amino acid residues of neuraminidase involved in influenza A virus entry. *Pathog. Dis.* **77**, ftz063 (2019).
8. McAuley, J. L., Gilbertson, B. P., Trifkovic, S., Brown, L. E. & McKimm-Breschkin, J. L. Influenza virus neuraminidase structure and functions. *Front. Microbiol.* **10**, 39 (2019).
9. Sriwilaijaroen, N. & Suzuki, Y. Molecular basis of a pandemic of avian-type influenza virus. *Methods Mol. Biol.* **1200**, 447–480 (2014).
10. Altman, M. O., Bennink, J. R., Yewdell, J. W., & Herrin, B. R. Lamprey VLRB response to influenza virus supports universal rules of immunogenicity and antigenicity. *Elife* **4**, e07467 (2015).
11. Rogers, G. N. & Paulson, J. C. Receptor determinants of human and animal influenza virus isolates: differences in receptor specificity of the H3 hemagglutinin based on species of origin. *Virology* **127**, 361–373 (1983).

12. Chandrasekaran, A. et al. Glycan topology determines human adaptation of avian H5N1 virus hemagglutinin. *Nat. Biotechnol.* **26**, 107–113 (2008).
13. Eisen, M. B., Sabesan, S., Skehel, J. J. & Wiley, D. C. Binding of the influenza A virus to cell-surface receptors: structures of five hemagglutinin-sialyloligosaccharide complexes determined by X-ray crystallography. *Virology* **232**, 19–31 (1997).
14. Shinya, K. et al. Avian flu: influenza virus receptors in the human airway. *Nature* **440**, 435–436 (2006).
15. Herfst, S. et al. Airborne transmission of influenza A/H5N1 virus between ferrets. *Science* **336**, 1534–1541 (2012).
16. Imai, M. et al. Experimental adaptation of an influenza H5 HA confers respiratory droplet transmission to a reassortant H5 HA/H1N1 virus in ferrets. *Nature* **486**, 420–428 (2012).
17. Xu, H. & Shaw, D. E. A simple model of multivalent adhesion and its application to influenza infection. *Biophys. J.* **110**, 218–233 (2016).
18. Lu, W. et al. Enhanced inhibition of influenza A virus adhesion by di- and trivalent hemagglutinin inhibitors. *J. Med. Chem.* **62**, 6398–6404 (2019).
19. Suptawiwat, O. et al. The N-linked glycosylation site at position 158 on the head of hemagglutinin and the virulence of H5N1 avian influenza virus in mice. *Arch. Virol.* **160**, 409–415 (2015).
20. Stevens, J. et al. Structure of the uncleaved human H1 hemagglutinin from the extinct 1918 influenza virus. *Science* **303**, 1866–1870 (2004).
21. Stevens, J. et al. Structure and receptor specificity of the hemagglutinin from an H5N1 influenza virus. *Science* **312**, 404–410 (2006).
22. Zhu, X. et al. A unique and conserved neutralization epitope in H5N1 influenza viruses identified by an antibody against the A/Goose/Guangdong/1/96 hemagglutinin. *J. Virol.* **87**, 12619–12635 (2013).
23. York, W. S. et al. GlyGen: computational and informatics resources for glycoscience. *Glycobiology* **30**, 72–73 (2020).
24. Xiong, X. et al. Enhanced human receptor binding by H5 haemagglutinins. *Virology* **456–457**, 179–187 (2014).
25. Pyankova, O. G. et al. Isolation of clade 2.3.4.4b A(H5N8), a highly pathogenic avian influenza virus, from a worker during an outbreak on a poultry farm, Russia, December 2020. *Euro Surveill.* **26**, 2100439 (2021).
26. Hiono, T. et al. Amino acid residues at positions 222 and 227 of the hemagglutinin together with the neuraminidase determine binding of H5 avian influenza viruses to sialyl Lewis X. *Arch. Virol.* **161**, 307–316 (2016).
27. Guo, H. et al. Highly pathogenic influenza A(H5Nx) viruses with altered H5 receptor-binding specificity. *Emerg. Infect. Dis.* **23**, 220–231 (2017).
28. Blumenkrantz, D., Roberts, K. L., Shelton, H., Lycett, S. & Barclay, W. S. The short stalk length of highly pathogenic avian influenza H5N1 virus neuraminidase limits transmission of pandemic H1N1 virus in ferrets. *J. Virol.* **87**, 10539–10551 (2013).
29. Yang, H., Carney, P. J., Chang, J. C., Villanueva, J. M. & Stevens, J. Structure and receptor binding preferences of recombinant hemagglutinins from avian and human H6 and H10 influenza A virus subtypes. *J. Virol.* **89**, 4612–4623 (2015).
30. Shi, Y. et al. Structures and receptor binding of hemagglutinins from human-infecting H7N9 influenza viruses. *Science* **342**, 243–247 (2013).
31. de Vries, E., Du, W., Guo, H. & de Haan, C. A. M. Influenza A virus hemagglutinin-neuraminidase-receptor balance: preserving virus motility. *Trends Microbiol.* **28**, 57–67 (2020).
32. Mammen, M., Choi, S.-K. & Whitesides, G. M. Polyvalent interactions in biological systems: implications for design and use of multivalent ligands and inhibitors. *Angew. Chem. Int. Ed. Engl.* **37**, 2754–2794 (1998).
33. de Vries, R. P. et al. Hemagglutinin receptor specificity and structural analyses of respiratory droplet-transmissible H5N1 viruses. *J. Virol.* **88**, 768–773 (2014).
34. Li, J., Zu Dohna, H., Cardona, C. J., Miller, J. & Carpenter, T. E. Emergence and genetic variation of neuraminidase stalk deletions in avian influenza viruses. *PLoS ONE* **6**, e14722 (2011).
35. Linster, M. et al. Identification, characterization, and natural selection of mutations driving airborne transmission of A/H5N1 virus. *Cell* **157**, 329–339 (2014).
36. Stevens, J. et al. Recent avian H5N1 viruses exhibit increased propensity for acquiring human receptor specificity. *J. Mol. Biol.* **381**, 1382–1394 (2008).
37. Zhang, W. et al. An airborne transmissible avian influenza H5 hemagglutinin seen at the atomic level. *Science* **340**, 1463–1467 (2013).
38. Gamblin, S. J. et al. The structure and receptor binding properties of the 1918 influenza hemagglutinin. *Science* **303**, 1838–1842 (2004).
39. Zhu, X. et al. Structural basis for a switch in receptor binding specificity of two H5N1 hemagglutinin mutants. *Cell Rep.* **13**, 1683–1691 (2015).
40. Hiono, T. et al. A chicken influenza virus recognizes fucosylated alpha2,3 sialoglycan receptors on the epithelial cells lining upper respiratory tracts of chickens. *Virology* **456–457**, 131–138 (2014).
41. Jang, S. G. et al. HA N193D substitution in the HPAI H5N1 virus alters receptor binding affinity and enhances virulence in mammalian hosts. *Emerg. Microbes Infect.* **13**, 2302854 (2024).
42. Peng, W. et al. Enhanced human-type receptor binding by ferret-transmissible H5N1 with a K193T mutation. *J. Virol*. **92**, e02016-17 (2018).
43. Liu, J. et al. Structures of receptor complexes formed by hemagglutinins from the Asian Influenza pandemic of 1957. *Proc. Natl Acad. Sci. USA* **106**, 17175–17180 (2009).
44. Lin, Y. P. et al. Evolution of the receptor binding properties of the influenza A(H3N2) hemagglutinin. *Proc. Natl Acad. Sci. USA* **109**, 21474–21479 (2012).
45. Xiong, X. et al. Receptor binding by a ferret-transmissible H5 avian influenza virus. *Nature* **497**, 392–396 (2013).
46. Chen, J. R. et al. Vaccination of monoglycosylated hemagglutinin induces cross-strain protection against influenza virus infections. *Proc. Natl Acad. Sci. USA* **111**, 2476–2481 (2014).
47. Mackay, R. M. et al. Airway surfactant protein D deficiency in adults with severe asthma. *Chest* **149**, 1165–1172 (2016).
48. Beresford, M. W. & Shaw, N. J. Bronchoalveolar lavage surfactant protein A, B, and D concentrations in preterm infants ventilated for respiratory distress syndrome receiving natural and synthetic surfactants. *Pediatr. Res.* **53**, 663–670 (2003).
49. Hartshorn, K. L. et al. Evidence for a protective role of pulmonary surfactant protein D (SP-D) against influenza A viruses. *J. Clin. Invest.* **94**, 311–319 (1994).
50. Hartley, C. A., Jackson, D. C. & Anders, E. M. Two distinct serum mannose-binding lectins function as beta inhibitors of influenza virus: identification of bovine serum beta inhibitor as conglutinin. *J. Virol.* **66**, 4358–4363 (1992).
51. Anders, E. M., Hartley, C. A. & Jackson, D. C. Bovine and mouse serum beta inhibitors of influenza A viruses are mannose-binding lectins. *Proc. Natl Acad. Sci. USA* **87**, 4485–4489 (1990).
52. Crouch, E. et al. Mutagenesis of surfactant protein D informed by evolution and x-ray crystallography enhances defenses against influenza A virus in vivo. *J. Biol. Chem.* **286**, 40681–40692 (2011).
53. An, Y., McCullers, J. A., Alymova, I., Parsons, L. M. & Cipollo, J. F. Glycosylation analysis of engineered H3N2 influenza A virus hemagglutinins with sequentially added historically relevant glycosylation sites. *J. Proteome Res.* **14**, 3957–3969 (2015).
54. An, Y. et al. *N*-glycosylation of seasonal influenza vaccine hemagglutinins: implication for potency testing and immune processing. *J. Virol.* **93**, e01693-18 (2019).
55. Parsons, L. M. et al. Influenza virus hemagglutinins H2, H5, H6, and H11 are not targets of pulmonary surfactant protein D: *N*-glycan subtypes in host-pathogen interactions. *J. Virol.* **94**, e01951–19 (2020).
56. Zeng, Z. et al. Characterization and evolutionary analysis of a novel H3N2 influenza A virus glycosylation motif in southern China. *Front. Microbiol.* **11**, 1318 (2020).
57. Li, B., Zhu, D., Shi, H. & Zhang, X. Effect of charge on protein preferred orientation at the air–water interface in cryo-electron microscopy. *J. Struct. Biol.* **213**, 107783 (2021).

58. Mastronarde, D. N. Automated electron microscope tomography using robust prediction of specimen movements. *J. Struct. Biol.* **152**, 36–51 (2005).
59. Lei, J. L. & Frank, J. Automated acquisition of cryo-electron micrographs for single particle reconstruction on an FEI Tecnai electron microscope. *J. Struct. Biol.* **150**, 69–80 (2005).
60. Zheng, S. Q. et al. MotionCor2: anisotropic correction of beam-induced motion for improved cryo-electron microscopy. *Nat. Methods* **14**, 331–332 (2017).
61. Zhang, K. Gctf: Real-time CTF determination and correction. *J. Struct. Biol*. **193**, 1–12 (2016).
62. Zivanov, J. et al. New tools for automated high-resolution cryo-EM structure determination in RELION-3. *Elife* 42166, https://doi.org/10.7554/eLife (2018).
63. Punjani, A., Rubinstein, J. L., Fleet, D. J. & Brubaker, M. A. cryoSPARC: algorithms for rapid unsupervised cryo-EM structure determination. *Nat. Methods* **14**, 290–296 (2017).
64. Pettersen, E. F. et al. UCSF Chimera–a visualization system for exploratory research and analysis. *J. Comput. Chem.* **25**, 1605–1612 (2004).
65. Emsley, P. & Cowtan, K. Coot: model-building tools for molecular graphics. *Acta Crystallogr. D Biol. Crystallogr.* **60**, 2126–2132 (2004).
66. Adams, P. D. et al. PHENIX: a comprehensive Python-based system for macromolecular structure solution. *Acta Crystallogr. D Biol. Crystallogr.* **66**, 213–221 (2010).
67. Potterton, L. et al. CCP4i2: the new graphical user interface to the CCP4 program suite. *Acta Crystallogr. D Struct. Biol.* **74**, 68–84 (2018).
68. Goddard, T. D. et al. UCSF ChimeraX: meeting modern challenges in visualization and analysis. *Protein Sci.* **27**, 14–25 (2018).

Acknowledgements

We thank the Tsinghua University Branch of the China National Center for Protein Sciences (Beijing), the Cryo-Electron Microscopy Facility and the Core Facility of Liangzhu Laboratory at Zhejiang University for providing the facility support. We thank Dr. Jianlin Lei, Xiaomin Li, and Fan Yang for cryo-EM data collection. We thank Dr. Haiteng Deng and Meng Han in the Proteinomics Facility at the Technology Center for Protein Sciences, Tsinghua University, for MS analysis of protein glycosylation. We thank Dr. Xinzheng Zhang and Dr. Wenbao Qi for helpful discussions. This work was supported by the National Key R&D Program of China (grants: 2023YFC2306300 and 2021YFA1300204), the National Natural Science Foundation of China (NSFC, grants: 31925023, 21827810, 31861143027), the Beijing Frontier Research Center for Biological Structure, the SXMU-Tsinghua Collaborative Innovation Center for Frontier Medicine and the Tsinghua-Peking Center for Life Sciences to Y.X., and the Excellent Youth Science Fund (Overseas) of NSFC, the Zhejiang Provincial Natural Science Foundation (LZ24C050001) to M.G.

Author contributions

R.F.L., J.J.G., M.G. and Y.X. conceived the project. J.J.G., R.F.L., L.W. and M.G. performed the experiments, analyzed data and prepared the figures. Y.X. supervised the research, planned the experiments and analyzed the data. Y.X., R.F.L., J.J.G. and M.G. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s44298-024-00059-9.

Correspondence and requests for materials should be addressed to Miao Gui or Ye Xiang.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2024